

Python HW

You have one hour to work on this homework. Find one uninterrupted hour

- In a markdown cell write down what time you started the homework and the AI model you are using
- work for one hour or until done using your notes and other sources but not AI.
- After one hour ask AI to do each question.
- paste the AI answer in below your answer and run it also.
- you should have two clearly delineated code cells for each question with two sets of answers
- in a markdown cell below the two answers comment on the answers, differences, and what you have learned. (1-4 sentences).
- If you got the correct answer and didn't need AI help you can just put a comment in, not needed and that you completed it. But I am assuming you learned something.
- Your homework should look similar to this notebook. The only difference is your code won't be hidden. I have greyed out my code.
- Grading is not based on how well you coded but on how well you learned to code.

Some AI notes

- If you want AI to solve the equation you can upload the Harris pdf or the latex equations. make it do the same equation as you.
- We always want to plot using `fig,ax=plt.subplots()`
 - So tell AI use `fig,ax=plt.subplots()`
- there are some other python plotting programs that will add bells and whistles. We only want to use matplotlib for now.
 - tell AI to only use matplotlib if it tries something else
- if AI does something you don't understand ask it to explain why. This can be a function, syntax, anything. Make it work for you to learn. You need to learn as much of this as you can if you are going to be successful
- You can also tell AI to clean up and put all the code together in one cell to run in Jupyter if it jumps around or things get messy.

Start time:

AI Model:

HW Introduction

Today's homework is an extension of class. We are going to do more practice of best fit lines. You are going to make up a hypothesis, make up or get data, and fit a line to the data. You will fit the data two ways. Once with your math and once with linregress.

- Your data and hypothesis can be anything and does not need to be "real". you can be creative.
- Your data needs to be created or found by you.
- You could interview friends to get data points. You can search online. You can make it as fun as you want. You can make something up! - - But you need to state a hypothesis. Make sure to clearly state your hypothesis. Have fun with it.
- You need at least 5 data points that you generate from your ideas and hypotheses. Make it look good! This is not about the data. This is about the idea of presenting a hypothesis and fitting a line! make it fun! Be goofy!

- One of your figures needs a figure caption.

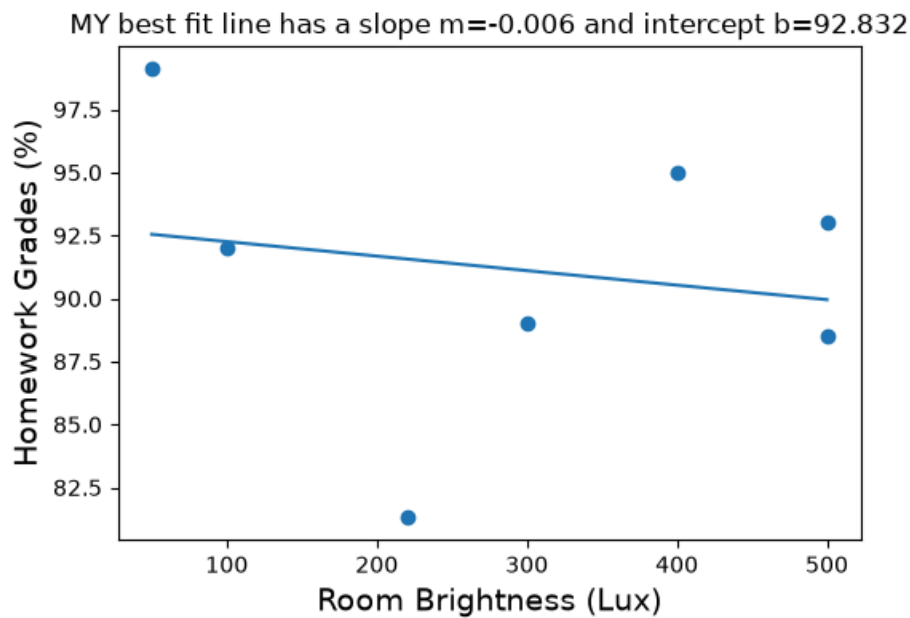
Hypothesis

Milstein 001 has been dark lately because of broken shades and lights. I hypothesize: The brighter that Milstein 001 is during class the higher the average Homework Grade

Question 1

Fit using your own Math and show the data and best fit line on a graph.

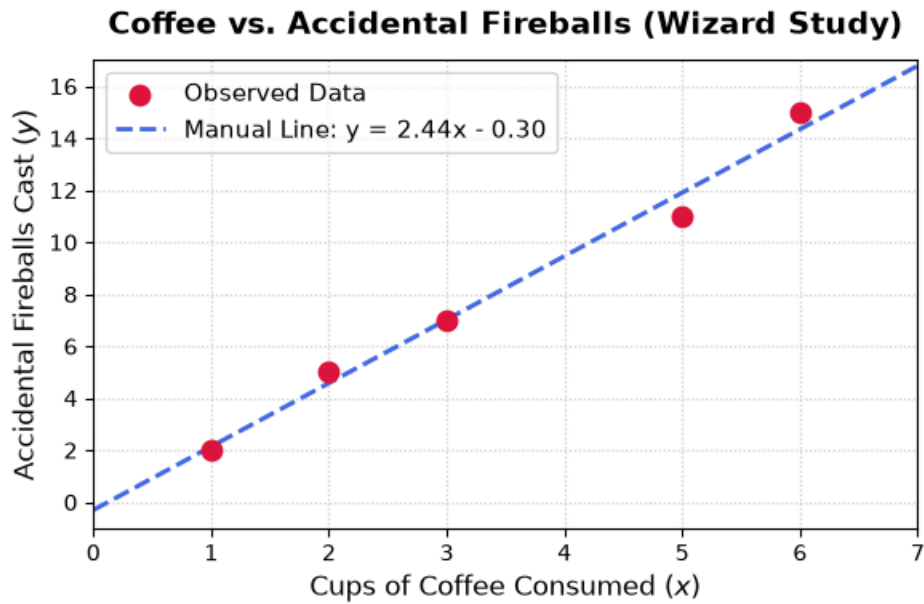
```
Out[28]: Text(0.5, 1.0, 'MY best fit line has a slope m=-0.006 and intercept b=92.832')
```



AI Answer

```
=== MANUAL FORMULA RESULTS ===
n = 5,  $\Sigma x = 17.0$ ,  $\Sigma y = 40.0$ ,  $\Sigma(x^2) = 75.0$ ,  $\Sigma(xy) = 178.0$ 
Manual Slope (m): 2.4419
Manual Intercept (b): -0.3023
Manual Equation:  $y = 2.4419x + (-0.3023)$ 

=== SCI PY LINREGRESS RESULTS ===
Linregress Slope (m): 2.4419
Linregress Intercept: -0.3023
```



Comparison

Q2

Repeat question 1 with the same data but instead use linregress and add a textbox to your graph that shows the values of slope, intercept, r-square, and p-value.

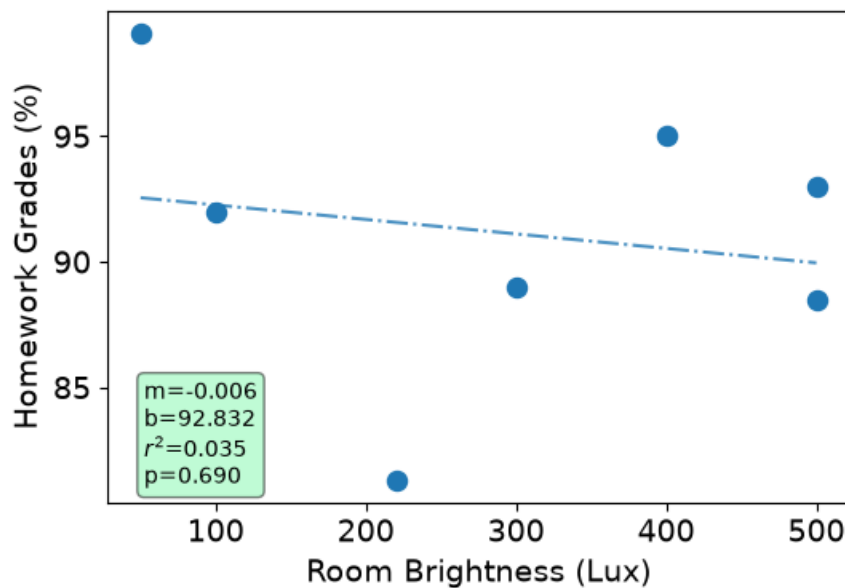
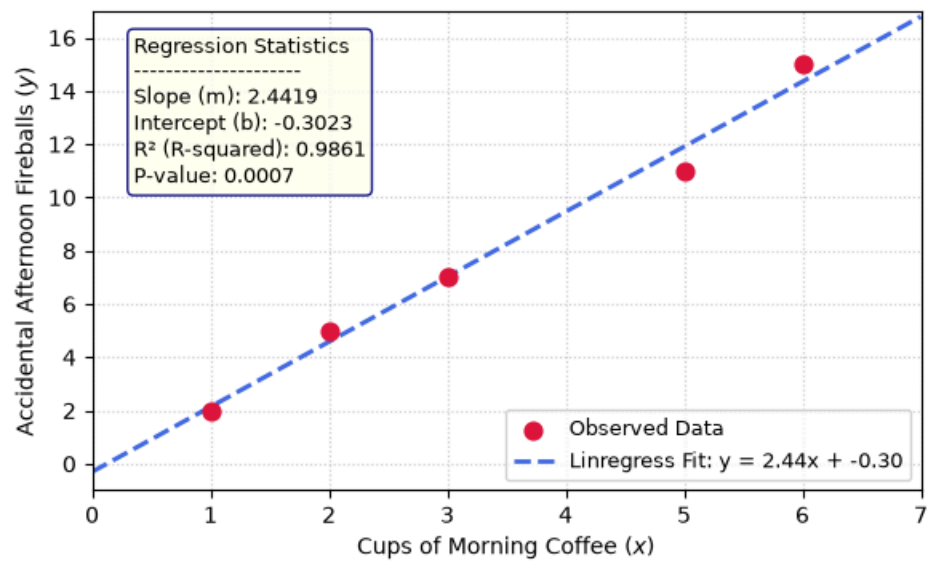


Figure Caption 1. Room Brightness versus average homework grades for students in Big Data with Python. There is no significant effect ($p > 0.05$) of room brightness in Milstein 001 on homework grades.

AI Answer

Statistical Analysis: Coffee vs. Wizard Fireballs



Comparison

